The empirical recurrence for OEIS A253656, recovered from its tail

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Abstract

OEIS A253656 records "Empirical recurrence of order 76" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 170$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 81$. Its last coefficient is $c_{76} = -576 \neq 0$, so no further tail satisfies anything shorter; any order-76 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A253656 is "Number of (n+1)X(2+1) 0..2 arrays with every 2X2 subblock diagonal maximum minus antidiagonal maximum unequal to its neighbors horizontally, vertically and nw-se diagonally.". It has offset 1 and begins

$$518, 3318, 25062, 181308, 1284808, 9172142, 65613880, 467877898, \ldots$$

The entry states:

> Empirical recurrence of order 76 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 14:08:39, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 170$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 170$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 76: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 76.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 340$ exact terms from $n = 5$ on, it returns order 76 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{76} c_i \, a(n-i),$$

c_1	0	c_{20}	1467368113	c_{39}	249389121126	c_{58}	−11269689711
c_2	36	c_{21}	−1247562014	c_{40}	−11545217824	c_{59}	−269158511174
c_3	210	c_{22}	−1341284627	c_{41}	−1337730066636	c_{60}	−117947309376
c_4	−98	c_{23}	−5684209066	c_{42}	634594804328	c_{61}	22497118568
c_5	−3434	c_{24}	6523195570	c_{43}	−688253337278	c_{62}	53605007042
c_6	−11537	c_{25}	12246987316	c_{44}	−219809538217	c_{63}	18721071876
c_7	4274	c_{26}	12985046204	c_{45}	348112918048	c_{64}	−3847740653
c_8	117419	c_{27}	−35934903996	c_{46}	3370442063194	c_{65}	−6171750518
c_9	283184	c_{28}	−43518663236	c_{47}	542696869850	c_{66}	−2822664054
c_{10}	17857	c_{29}	963779812	c_{48}	−3298591327020	c_{67}	25390326
c_{11}	−2214736	c_{30}	106021182457	c_{49}	−4104450215898	c_{68}	410807157
c_{12}	−3991080	c_{31}	−7197851330	c_{50}	818917306491	c_{69}	236507754
c_{13}	−638126	c_{32}	25592607439	c_{51}	4058527415096	c_{70}	100524893
c_{14}	31099986	c_{33}	−188545497186	c_{52}	2199713248436	c_{71}	−6557958
c_{15}	24663750	c_{34}	245600510694	c_{53}	−1185599467326	c_{72}	−8784645
c_{16}	−13237501	c_{35}	−226973046472	c_{54}	−2216362586524	c_{73}	−6194040
c_{17}	−288029042	c_{36}	28308301135	c_{55}	−560370241834	c_{74}	−1992912
c_{18}	52019305	c_{37}	−394767843388	c_{56}	805165469465	c_{75}	−211392
c_{19}	201023910	c_{38}	794079516017	c_{57}	648492031618	c_{76}	−576

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 81$, and it is the recurrence OEIS A253656 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{76} - c_1 x^{76-1} - \cdots - c_{76}$. Its constant term is $-c_{76} = 576$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 76 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 76, so $\deg q = 76$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 76, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 76, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -576.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-76 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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